CA-ResNet: A Deep Learning-Based Framework for Uplink Signal Detection of LoRa-Based LEO Satellite IoT

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***Abstract*—This letter proposes a novel deep learning based framework for uplink signal detection of Long-Range (LoRa) based low Earth orbit (LEO) satellite Internet of Things (IoT). First, a spherical stochastic geometry based analytical framework is developed, where the terrestrial LoRa end-devices are modeled through a Poisson point process within the satellite coverage. Then, a complex-aware residual network (CA-ResNet) is proposed, which fully leverages the power differences, as well as the relative time and frequency offsets between the desired and interfering signals to achieve implicit interference cancellation. Simulation results demonstrate that the proposed scheme outperforms the conventional counterparts.**

***Index Terms*—Deep learning, Internet of Things (IoT), LoRa, low Earth orbit (LEO) satellite, signal detection.**

I. Introduction

FACILITATED by the chirp spread spectrum (CSS) modulation used in physical (PHY) layer, Long-Range (LoRa) can achieve a flexible trade-off between coverage and data rate, thereby accommodating heterogeneous low Earth orbit (LEO) satellite Internet of Things (IoT) applications [1], [2]. However, due to the exponentially growing number of IoT end-devices (EDs), along with the ALOHA-based uncoordinated transmissions of LoRa medium access control (MAC) layer, the traditional non-coherent detection faces a significant bit error rate (BER) performance degradation in interference-limited scenarios [3]. In this context, exploiting deep learning (DL) based methods for interference mitigation and signal detection has been envisioned as a promising approach.

In this field, the initial study began with a convolutional neural network (CNN) based detector proposed in [4], which aimed to deal with additive white Gaussian noise (AWGN) together with time offset (TO) and frequency offset (FO). Moreover, to cope with the inter-ED interference, two DL-based strategies were proposed in [5], namely, regression for bit detection based on deep neural network (DNN) and classification for symbol detection based on CNN. To further improve the scalability of LoRa networks, the authors in [6] proposed a LoRa signal detector based on an autoencoder for denoising, followed by a CNN for symbol detection. In addition, to handle the signal detection under the impact of both channel fading and inter-ED interference, the authors in [7] proposed a CNN-based joint channel estimation and signal detection scheme, which can achieve better BER performance compared to the classical detectors. However, the aforementioned works merely focused on the signal detection of terrestrial LoRa networks, while few studies investigate the uplink signal detection of LoRa-based LEO satellite IoT. Given that the low signal detection accuracy in the presence of inter-ED interference has become a critical bottleneck for such a new paradigm, we provide a novel DL-based framework to further improve the BER performance in interference-limited scenarios. The main contributions of this letter are summarized as follows.

1) To characterize the practical network deployment of LoRa-based LEO satellite IoT, a new spherical stochastic geometry (SG) based analytical framework is developed, where the terrestrial LoRa EDs are modeled through a Poisson point process (PPP) on the Earth surface within the satellite coverage. Based on this, the uplink received signal model, containing the desired signal, cumulative interfering signals, and noise, is obtained according to the random asynchronous access characteristic of LoRa MAC layer.
2) To further improve the signal detection accuracy in interference-limited scenarios, a complex-aware residual network (CA-ResNet) is proposed based on the domain knowledge of LoRa PHY layer, which fully leverages the power differences, the relative TOs, as well as the relative FOs between the desired and interfering signals to achieve implicit interference cancellation while improving the BER performance. Simulation results demonstrate that the proposed scheme outperforms the conventional counterparts.

Notations: Throughout this letter, vectors and matrices are represented by lower-case and upper-case boldface letters, respectively. $\odot$, $(\cdot)^*$, and $(\cdot)^T$ denote the Hadamard product, conjugate, and transpose operations. $\max\{\cdot\}$, $\min(\cdot)$, $\lceil\cdot\rceil$, $|\cdot|$, $\Re\{\cdot\}$, and $\Im\{\cdot\}$ are the maximum, minimum, ceiling, absolute value, real part, and imaginary part operations. $\mathcal{U}(\cdot,\cdot)$ denotes the uniform distribution. $\mathcal{N}(\cdot,\cdot)$ and $\mathcal{CN}(\cdot,\cdot)$ represent the real and complex Gaussian distributions, respectively.

II. LoRa Modulation Basics

CSS modulation is adopted in LoRa PHY layer with two important parameters, i.e., bandwidth B and spreading factor (SF) k. Under Nyquist sampling, each chirp is represented by M complex-valued baseband samples, where $M = 2^k$ with

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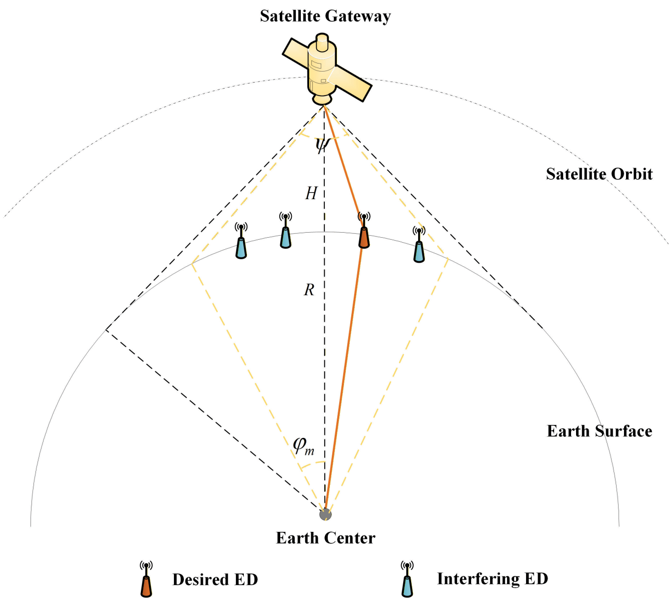


Fig. 1. Direct communications between the terrestrial LoRa EDs and the LEO satellite.

$k \in \{7, \ldots, 12\}$. In particular, the basic upchirp is defined as [8]

$$u[n] = \exp\left\{ j2\pi \left(\frac{n^2}{2M} - \frac{n}{2} \right) \right\}, \tag{1}$$

where $n = 0, 1, \ldots, M-1$.

For LoRa modulation, each group of k information bits $b_0, b_1, \ldots, b_{k-1}$ is first mapped to a data symbol, denoted as $m = \sum_{i=0}^{k-1} b_i 2^i \in \mathcal{L} = \{0, 1, \ldots, M-1\}$. Then, the corresponding modulated chirp symbol is expressed as

$$s_m[n] = u[n] \exp\left\{ j\frac{2\pi mn}{M} \right\}. \tag{2}$$

The received signal in an AWGN channel is given by

$$r[n] = s_m[n] + w[n], \tag{3}$$

with $w[n] \sim \mathcal{CN}\left(0, \sigma^2\right)$. Note that $\sigma^2 = -174 + N_F + 10\log_{10} B$, where N_F represents the receiver's noise figure.

The traditional non-coherent detection consists of three main steps: dechirp, discrete Fourier transform (DFT), and frequency domain (FD) peak index detection [8]. Specifically, let $\mathbf{r} = [r[0], \ldots, r[M-1]]^T$ and $\mathbf{u} = [u[0], \ldots, u[M-1]]^T$, the DFT output can be expressed as

$$\mathbf{y} = \mathbf{F}_M (\mathbf{r} \odot \mathbf{u}), \tag{4}$$

where $\mathbf{F}_M \in \mathbb{C}^{M \times M}$ denotes the M-point DFT matrix with entries $\mathbf{F}_M(l, n) = \exp\left(-j\frac{2\pi ln}{M}\right)$.

Finally, the data symbol is obtained as

$$\hat{m} = \arg\max_{l \in \mathcal{L}} \{|\mathbf{y}|\}. \tag{5}$$

III. System Model

A. Network Model

As illustrated in Fig. 1, the onboard gateway directly communicates with the terrestrial LoRa EDs within the satellite coverage. In particular, the contact angle φ is adopted to characterize the geometric relationship between a LoRa ED and the LEO satellite. Given the satellite effective beamwidth ψ, the maximum contact angle of the LoRa ED can be expressed as [9]

$$\varphi_m = \begin{cases} \sin^{-1}\left(\frac{R+H}{R} \sin\left(\frac{\psi}{2}\right) \right) - \frac{\psi}{2}, & \psi < \psi_0, \\ \cos^{-1}\left(\frac{R}{R+H} \right), & \psi \geq \psi_0, \end{cases} \tag{6}$$

where R is the radius of Earth, H is the orbit altitude, and $\psi_0 = 2\sin^{-1}\left(\frac{R}{R+H}\right)$. Given the contact angle φ, the contact distance between a LoRa ED and the satellite can be obtained as

$$d(\varphi) = \sqrt{(R+H)^2 + R^2 - 2R(R+H)\cos\varphi}. \tag{7}$$

Without loss of generality, the terrestrial LoRa EDs are randomly and uniformly distributed on the Earth surface within the satellite coverage, modeled as a PPP Φ of intensity λ, while the satellite coverage area can be expressed as $|\mathcal{R}| = 2\pi R^2 (1 - \cos\varphi_m)$. Assume that the SF allocation ratio is denoted by ρ_k, and thus class k LoRa EDs are modeled as an independent PPP Φ_k of intensity $\lambda_k = \rho_k \lambda$.

Furthermore, let Φ_k^A denote the class k active EDs, modeled as a PPP of intensity $p_k \lambda_k$ with $p_k = \frac{T_k}{T_p}$ representing the active probability of class k EDs. Herein, T_p denotes the average packet inter-arrival time while T_k is the Time-on-Air (ToA) of class k EDs, given by [10]

$$T_k = (N_P + 4.25 + N_L)\frac{2^k}{B}, \tag{8}$$

where N_P is the number of upchirp symbols in the preamble while N_L is the number of modulated chirp symbols in the PHY payload.[1] Notably, N_P is usually set to a default value, while N_L can be further expressed as

$$N_L = \max\left\{ \left\lceil \frac{8L - 4k + 28 + 16C - 20H_E}{4k} \right\rceil (C_R + 4), 0 \right\} + 8, \tag{9}$$

where L is the payload length calculated in bytes, $H_E = 0$ represents an explicit header, $H_E = 1$ represents an implicit header, $C = 0$ represents no cyclic redundancy check (CRC), $C = 1$ represents CRC appending, $C_R \in \{1, 2, 3, 4\}$ determines the coding rate.

B. Channel Model

In general, both large-scale and small-scale fading should be considered in the channel model. Various generalized composite fading models have been developed to characterize the complex propagation environment, such as λ-κ-μ and α-η-μ/IGA fading distributions [11], [12]. However, since the LoRa EDs are often distributed in open remote areas for LEO satellite IoT scenarios, only large-scale fading is considered to characterize the propagation environment while simplifying the subsequent analysis. Specifically, a newly developed

[1] It should be noted that LoRa-based IoT communications are generally delay-tolerant short-packet transmissions with low radio duty cycle and low traffic volume. Therefore, ALOHA-based random access is usually adopted in the MAC layer without scheduled time slot allocation [10].

empirically-verified channel model is adopted to characterize the large-scale fading of satellite-to-ground radio links [13].

On one hand, the free-space path-loss (FSPL) can be expressed as

$$L(\varphi) = \left(\frac{c}{4\pi f_c d(\varphi)}\right)^2, \tag{10}$$

where c and f_c denote the velocity of light and the carrier frequency, respectively.

On the other hand, benefited from the line-of-sight (LoS) and non-line-of-sight (NLoS) probability formulations, the excess path-loss (EPL) follows a mixed Gaussian distribution, which can be expressed as [13]

$$\begin{aligned}\eta\,[\mathrm{dB}] \sim\; & p_{LoS}(\varphi)\mathcal{N}\left(-\mu_{LoS},\sigma_{LoS}^2\right)\\ & + (1-p_{LoS}(\varphi))\mathcal{N}\left(-\mu_{NLoS},\sigma_{NLoS}^2\right),\end{aligned} \tag{11}$$

where $p_{LoS}(\varphi) = \exp\left(-\frac{\beta\sin\varphi}{\cos\varphi-\alpha}\right)$ represents the probability of LoS propagation, $\alpha = \frac{R}{R+H}$, while parameters $\beta, \mu_{LoS}, \sigma_{LoS}, \mu_{NLoS}, \sigma_{NLoS}$ depend on the specific propagation environment. Moreover, given the ED's contact angle, the EPL can be calculated using its mean value, given by

$$\begin{aligned}\bar{\eta}(\varphi) = \;& p_{LoS}(\varphi)\exp\left(\frac{\zeta^2\sigma_{LoS}^2}{2} - \zeta\mu_{LoS}\right)\\ & + (1-p_{LoS}(\varphi))\exp\left(\frac{\zeta^2\sigma_{NLoS}^2}{2} - \zeta\mu_{NLoS}\right),\end{aligned} \tag{12}$$

where $\zeta \triangleq \frac{\ln(10)}{10}$.

C. Signal Model

Owing to the quasi-orthogonality of different SFs, we can only consider the impact of inter-ED interference within each class of LoRa EDs [10]. Let the desired and interfering EDs be characterized by their contact angles φ_o and φ_i, transmit powers P_o and P_i, antenna gains G_o and G_i, respectively. Therefore, the uplink received signal is given by

$$\begin{aligned} r[n] = \;& \underbrace{\sqrt{P_oG_oG_sL(\varphi_o)\bar{\eta}(\varphi_o)}s_m[n]}_{\text{desired signal}} + \underbrace{w[n]}_{\text{noise}}\\ & + \underbrace{\sum_{i\in\Phi_k^A\backslash o}\sqrt{P_iG_iG_sL(\varphi_i)\bar{\eta}(\varphi_i)}s_{I_i}[n]\exp\left\{j\frac{2\pi\epsilon_i n}{M}\right\}}_{\text{cumulative interfering signals}}\\ = \;& J_os_m[n] + \sum_{i=1}^{N_I}J_is_{I_i}[n]\exp\left\{j\frac{2\pi\epsilon_i n}{M}\right\} + w[n], \end{aligned} \tag{13}$$

where G_s denotes the satellite antenna gain, N_I denotes the number of interfering EDs which follows a Poisson distribution with parameter $p_k\rho_k\lambda|\mathcal{R}|$, $J_o = \sqrt{P_oG_oG_sL(\varphi_o)\bar{\eta}(\varphi_o)}$, $J_i = \sqrt{P_iG_iG_sL(\varphi_i)\bar{\eta}(\varphi_i)}$, $s_m[n]$ and $s_{I_i}[n]$ represent the unit-power desired signal and i-th interfering signal, respectively, and $\epsilon_i \sim \mathcal{U}(-0.5, 0.5)$ denotes the normalized relative FO between the desired signal and i-th interfering signal with respect to $\Delta f = \frac{B}{M}$. Notably, both laboratory and in-orbit satellite experiments have demonstrated the robustness of LoRa technology against the Doppler shift [14]. Therefore, benefited from the efficient time-frequency synchronization algorithms employed in LoRa PHY layer [15], we can assume that the satellite gateway is synchronized to the desired signal [5], [6], [7]. However, due to the distinct FOs experienced by different EDs, relative FOs that consist of both integer and fractional FOs may arise between the desired signal and interfering signals. As indicated in [16], the integer FO only induce a cyclic shift of the interference peak index in the DFT output without causing energy dispersion. Hence, only the impact of fractional FO is taken into consideration, i.e., $\epsilon_i \in [-0.5, 0.5]$ for $i \in \{1,\ldots,N_I\}$, which leads to the energy dispersion of the interference peak in the DFT output.

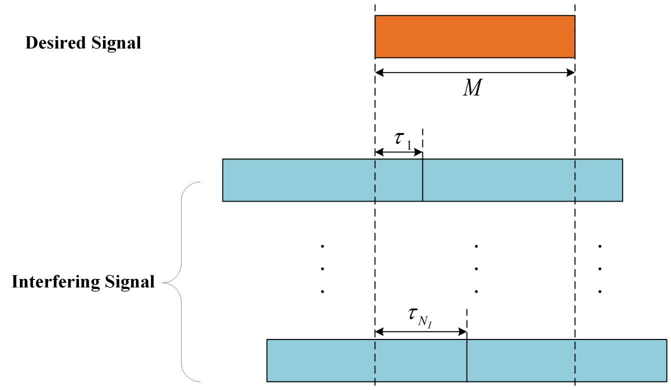


Fig. 2. Uplink received signal model.

In addition to the relative FOs, random asynchronous transmission will inevitably introduce relative TOs between the desired signal and interfering signals, as shown in Fig. 2. Due to the existence of relative TO $\tau_i \sim \mathcal{U}(0, M-1)$, the corresponding interfering signal $s_{I_i}[n]$ for $i \in \{1,\ldots,N_I\}$ consists of two parts from different chirp symbols within the duration of the desired signal. Specifically, the i-th interfering signal $s_{I_i}[n]$ can be expressed as

$$s_{I_i}[n] = \begin{cases} s_{i_1}[n], & 0 \le n < \tau,\\ s_{i_2}[n], & \tau \le n < M-1, \end{cases} \tag{14}$$

where i_1, i_2 are the corresponding data symbols of the i-th interfering signal.

IV. Proposed CA-ResNet

A. Input Pre-Processing

First, the time domain received signal, containing the desired signal, cumulative interfering signals, and noise, is transformed into FD via dechirp and DFT operations, which is identical to the traditional non-coherent detection. More specifically, let $\mathbf{r} = [r[0],\ldots,r[M-1]]^T$ and $\mathbf{u} = [u[0],\ldots,u[M-1]]^T$, the DFT output is obtained as $\mathbf{y} = \mathbf{F}_M(\mathbf{r}\odot\mathbf{u}) \in \mathbb{C}^M$. Then, the amplitude, real part, and imaginary part of $\mathbf{y}$ are normalized, respectively, expressed as

$$\mathbf{y}^1 = \frac{|\mathbf{y}| - \min\{|\mathbf{y}|\}}{\max\{|\mathbf{y}|\} - \min\{|\mathbf{y}|\} + \delta}, \tag{15}$$

$$\mathbf{y}^2 = \frac{\Re\{\mathbf{y}\}}{\max\{|\mathbf{y}|\} + \delta}, \tag{16}$$

$$\mathbf{y}^3 = \frac{\Im\{\mathbf{y}\}}{\max\{|\mathbf{y}|\} + \delta}, \tag{17}$$

where δ is a constant much smaller than the variance of $|\mathbf{y}|$.

TABLE I
NETWORK STRUCTURE OF CA-RESNET

Layer	Size	Activation
Input Layer	$1 \times M \times 3$	-
1D Convolutional Layer	32, 1×7	-
Batch Normalization Layer	-	ReLU
Residual Block 1	-	-
1D Convolutional Layer	32, 1×5	-
Batch Normalization Layer	-	ReLU
1D Convolutional Layer	32, 1×5	-
Batch Normalization Layer	-	Shortcut + ReLU
Residual Block 2	-	-
1D Convolutional Layer	64, 1×5	-
Batch Normalization Layer	-	ReLU
1D Convolutional Layer	64, 1×5	-
Batch Normalization Layer	-	-
1D Convolutional Layer	64, 1×1	Shortcut + ReLU
Dropout Layer	0.4	-
1D Convolutional Layer	1, 1×1	-
Output Layer	M	Softmax

TABLE II
SYSTEM PARAMETERS

Parameter	Value
Carrier frequency f_c	868 MHz
Bandwidth B	125 kHz
Earth radius R	6371 km
Orbit altitude H	500 km
PHY payload length L	50 bytes
Average packet inter-arrival time T_p	3600 s
Satellite effective beamwdith ψ	50°
Normalized relative FO ϵ_i	$\mathcal{U}(-0.5, 0.5)$
ED's antenna gain G_o, G_i	2.15 dBi
Interfering ED's transmit power P_i	14 dBm
Interfering ED's contact angle φ_i	$\mathcal{U}(0, \varphi_m)$
Noise figure N_F	6 dB
Satellite antenna gain G_s	10 dBi
LoS EPL mean μ_{LoS}	0 dB
LoS EPL standard deviation σ_{LoS}	2.8 dB
LoS probability parameter β	2.3
NLoS EPL mean μ_{NLoS}	12 dB
NLoS EPL standard deviation σ_{NLoS}	9 dB

B. Symbol Detection

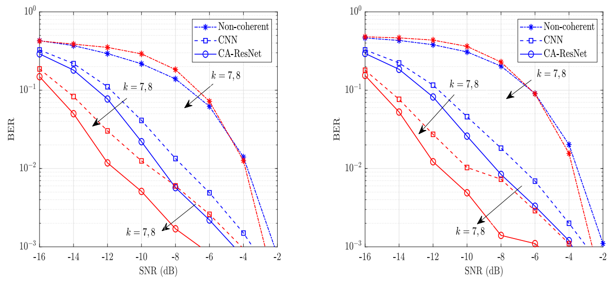
The pre-processed signal $\mathbf{Y} = \left[\mathbf{y}^1; \mathbf{y}^2; \mathbf{y}^3\right]^T$ serves as the input for CA-ResNet-based symbol detection module. Unlike the traditional non-coherent detection which only utilizes the amplitude information for FD peak index detection, the proposed CA-ResNet distinguishes the target peak from interference peaks by exploiting both the amplitude and phase information of the DFT output, enabling more accurate feature extraction and interference cancellation.

First, the three-channel FD signals are fed into a one-dimensional (1D) convolutional layer to perform preliminary feature extraction. Based on this, the core of CA-ResNet consists of two residual blocks containing multiple 1D convolutions, where the number of convolutional layers and the size of convolutional filters are detailed in Table I. The hyper-parameters of CA-ResNet, such as the number of convolutional layers and the number and size of filters in each convolutional layer, are determined via random search algorithm [17]. Next, a side-by-side discussion that clearly differentiates our proposed CA-ResNet from other related DL-based LoRa detectors is given as follows. First of all, raw time domain signal is used as input for the CNN-based detector in [4] while CA-ResNet utilizes dechirp and DFT operations for input-processing, which fully leverages the domain knowledge of LoRa PHY layer for feature extraction. Besides, by leveraging the identity mapping property of skip connections in residual blocks, CA-ResNet can be trained to specifically learn the residual features between the interference-limited and interference-free DFT outputs, thereby achieving implicit interference cancellation and better BER performance compared to the CNN-based detectors in [5] and [7]. Notably, to address the issue that fully connected layer could destroy the spatial topology of feature maps and lead to the loss of FD index information, the proposed CA-ResNet adopts a 1×1 fully convolutional design after the residual blocks, which strictly preserves the position information of FD index, thus accommodating the essence of LoRa modulation that utilizes different frequency bins to represent the data symbols. Moreover, compared to the cascaded network proposed in [6] which consists of an autoencoder for denoising and a CNN for symbol detection, CA-ResNet achieves a much lower computational complexity which can be expressed as $\mathcal{O}\left(M \log_2 M + M + \sum_{j=1}^{J} DK_{1j}K_{2j}M\right)$, where J denotes the number of convolutional layers, D represents the kernel size, K_{1j} and K_{2j} represent the number of input and output channels, respectively. By contrast, the overall computational complexity of the cascaded autoencoder-CNN network is approximated as $\mathcal{O}\left(M^3\right)$ [6]. Therefore, our proposed CA-ResNet is a lightweight 1D residual network, which is more suitable for resource-constrained onboard signal processing.

V. SIMULATION RESULTS AND DISCUSSIONS

Unless otherwise stated, the system parameters adopted in numerical simulations are listed in Table II, and without loss of generality, random SF allocation is employed with $\rho_k = \frac{1}{6}$, $\forall k$. Besides, the LoRa PHY layer packet parameters are set as $N_P = 8$, $C = 1$, $H_E = 0$, and $C_R = 1$. During the training process of our proposed CA-ResNet, a mixed data set with the signal-to-noise ratio (SNR) value of the desired signal taken in the interval $[-16, 0]$ dB is generated, which characterizes the typical SNR range in LoRa networks [15]. In addition, the network is trained with 40 epochs and a batch size of 256 samples, where the adaptive moment estimation (ADAM) algorithm is adopted to optimize the network parameters with an initial learning rate of 0.001.

Herein, Fig. 3 presents the BER performance comparison between the traditional non-coherent detection, the CNN-based



(a) $\lambda = 1/\text{km}^2$ (b) $\lambda = 2/\text{km}^2$

Fig. 3. BER performance comparison between the traditional non-coherent detection, the CNN-based detector, and the proposed CA-ResNet for (a) $\lambda = 1/\text{km}^2$ and (b) $\lambda = 2/\text{km}^2$.

detector, and the proposed CA-ResNet, where the physical meaning of λ denotes the number of LoRa EDs per square kilometer. It can be observed that the proposed CA-ResNet consistently achieves the lowest BER across the considered SNR range, followed by the CNN-based detector, while the traditional non-coherent detection exhibits the worst error performance. This is because the traditional non-coherent detection suffers from decision errors due to the interference peaks in the DFT output especially under low SNR conditions. Moreover, since the proposed CA-ResNet can be trained to specifically learn the residual features between the interference-limited and interference-free DFT outputs, it can fully leverage the power differences, the relative time and frequency offsets between the desired and interfering signals to perform implicit interference cancellation, thus obtaining error performance gains compared to the CNN-based detector. Furthermore, the proposed scheme achieves significant BER performance improvements with the increase of SF k, whereas the traditional non-coherent detection yields limited performance gains due to the absence of robustness to inter-ED interference, which is consistent with the conclusions in [18]. Additionally, since the proposed CA-ResNet adopts the input pre-processing based on the traditional non-coherent detection, the trained network model exhibits good generalization capability and scalability. However, if key system parameters, like the ED's density λ, undergo substantial changes, the network architecture and model parameters of CA-ResNet need to be fine-tuned.

VI. Conclusion

A novel signal detection scheme for LoRa-based LEO satellite IoT has been proposed in this letter through a cross-fertilization between spherical SG and DL to handle the inter-ED interference. Specifically, a new spherical SG-based analytical framework has been developed to characterize the practical network deployment of LoRa-based LEO satellite IoT, and a tailored CA-ResNet has been proposed for interference mitigation and signal detection. Numerical simulations have shown the superiority of our proposed scheme compared to the traditional counterparts in interference-limited scenarios. Overall, we believe our work will inspire the future research of applying DL-based approach to further improve the signal detection accuracy of LoRa-based non-terrestrial networks (NTNs).

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